

AAGENT Smart Contract Security Assessment

AAGENT

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Table of Contents

Table of Contents	2
Summary	3
Scope	4
System Overview	5
Security Model and Trust Assumptions	6
Medium Severity · 4 resolved	7
Low Severity · 1 resolved	11
Notes & Additional Information · 4 resolved	12
Verification and Closure	16
Conclusion	36
Disclaimer	37

Summary

ALL TRACKED CODE FINDINGS VERIFIED AS RESOLVED

Assessment result

No Critical or High issue was identified. All tracked code findings have verified remediation in the current source snapshot. Mainnet activation remains subject to deployment-address, role, reserve, RPC and funded-wallet acceptance gates.

Scope

Five core contracts and six production interfaces, 3,287 production lines of Solidity, plus deployment and keeper controls.

Verification

34 Solidity files compiled; 74 chain-56 tests passed with one intentional wrong-chain case skipped by design; the chain-97 guard passed independently.

Scope

Included

Market configuration, phase pricing, purchases, redemption cost basis, reserves, oracle safety, staking custody, ownership and deployment controls.

Excluded

Third-party token implementations, live RPC availability, exchange listing, production key custody and any address not bound to the printed scope root.

Change policy

Any byte-level change to an in-scope file requires a new scope root and regression execution.

System Overview

Registry

Authoritative 90-day schedule, fixed 10/7/3 phase rates, market modes, endpoint precommitment and configuration versioning.

Swap and Redemption

Purchases validate payment facts, caps, inventory and reserve coverage; refunds use actual cost basis and one-time wallet nonces.

Vault and Staking

Sale proceeds remain locked and staking records full entitlement until the matured AGNT claim.

Security Model and Trust Assumptions

Safety

Execution fails closed on stale, inconsistent or unverifiable market facts.

Liveness

Redemption remains available independently of purchase-oracle health when the market mode permits.

External systems

Feeds, RPC services and deployment key custody are treated as explicit operational dependencies.

Privileged Roles

Owner

Configuration is owner-restricted and ownership cannot be renounced.

Keeper

Any keeper may persistently pause unhealthy purchase feeds but cannot disable redemption-only modes.

Beneficiary

Staking proceeds release only after a matured user claim.

MED-01 - Cumulative redemption threshold

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with cumulative redemption threshold could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

Wallet-level cumulative UTC-day accounting now evaluates the next total before the hard limit and manual threshold.

Validation

Targeted regression tests pass for split transactions, threshold equality and hard-limit enforcement.

MED-02 - Redemption reserve coverage

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with redemption reserve coverage could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

BNB, USDT and USDC purchase execution now fails closed unless the refund reserve covers projected cost-basis liability.

Validation

Fresh tests cover reserve readability, liability growth and insufficient-reserve rejection.

MED-03 - Persistent oracle circuit breaker

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with persistent oracle circuit breaker could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

Per-feed maximum age, absolute bounds, deviation limits and permissionless persistent purchase pause are implemented.

Validation

Breaker reason classification, healthy-feed rejection and post-launch redemption continuity tests pass.

MED-04 - Staking oracle and operational controls

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with staking oracle and operational controls could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

Staking validates round completeness, timestamps, decimals, asset/feed collisions, quote lifetime, slippage and inventory.

Validation

Constructor, quote, custody, reentrancy and keeper safety suites pass; production activation remains gated by deployment controls.

LOW-01 - Redemption rate-limit semantics

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with redemption rate-limit semantics could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

The enforced cumulative daily wallet limit and manual threshold prevent transaction splitting and bound wallet throughput.

Validation

Boundary tests pass; the chosen UTC-day policy is documented as the intended control.

INFO-01 - Event evidence completeness

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with event evidence completeness could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

Canonical transaction, event, nonce, position and block-hash evidence is persisted for settlement workflows.

Validation

Duplicate and drift conditions fail closed in the service verification layer.

INFO-02 - NatSpec and interface clarity

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with natspec and interface clarity could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

External interfaces and security-sensitive configuration fields are documented and version-bound.

Validation

Compilation succeeds across 34 Solidity files with the current interfaces.

INFO-03 - Deployment reproducibility

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with deployment reproducibility could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

The deployment script validates the exact 90-day schedule, chain 56, feed constraints, predicted addresses and constructor events.

Validation

Eight deployment-safety regression cases pass.

INFO-04 - Operational runbook controls

STATUS - RESOLVED

Status

RESOLVED

Risk

The earlier control gap associated with operational runbook controls could weaken enforcement under adversarial sequencing or dependency failure.

Remediation

Keeper configuration, irreversible broadcast confirmation and public-artifact secret stripping are encoded as fail-closed checks.

Validation

Keeper and deployment tests reject missing, malformed, testnet and secret-bearing inputs.

Finding Closure Matrix

9 OF 9 TRACKED FINDINGS RESOLVED

Medium

MED-01 through MED-04: RESOLVED.

Low

LOW-01: RESOLVED.

Informational

INFO-01 through INFO-04: RESOLVED.

Compilation Results

COMPILATION - PASS

Result

34 Solidity files compiled successfully with solc 0.8.24, optimizer 500 and EVM target paris.

Dependencies

OpenZeppelin Contracts 5.1.0 and pinned Hardhat plugins.

Test Results

TESTS - PASS

Chain 56

74 passing tests; one wrong-chain constructor case skipped by design.

Chain 97

Independent wrong-chain guard passed 1/1.

Coverage Results

Production

96.02% statements; 59.30% branches; 97.09% functions; 86.25% lines.

All instrumented

96.18% statements; 59.91% branches; 97.66% functions; 87.01% lines.

Static Analysis

STATIC ANALYSIS - COMPLETE

Tool

Slither 0.11.5 completed successfully.

Triage

71 raw detector records were manually classified against the AEGENT design.

Result

No unresolved Critical or High issue remained.

Dependency Audit

DEPENDENCY AUDIT - PASS

Production

npm audit --omit=dev reported zero known production dependency advisories.

Control

Security-sensitive versions are pinned in the frozen evidence package.

Deployment Script Safety

Chain

Requires BNB Chain mainnet chain 56 and exact irreversible broadcast confirmation.

Feeds

Validates sale window, phase boundaries, feed age, decimals and risk settings.

Artifacts

Rejects testnet inputs and strips signer and RPC secrets from public artifacts.

Wrong-Chain Protection

WRONG-CHAIN GUARD - PASS

Constructor

AagentStakingV1 refuses deployment outside chain 56.

Regression

The independent chain-97 path confirms rejection.

Operational Controls

Keeper

Fails closed on missing or malformed configuration.

Pause semantics

Purchase feed failures cannot disable redemption-only or post-launch redemption.

Roles

Pause and withdrawal invariants are explicitly tested.

Regression Matrix

Accounting

Cost basis, partial refunds, liability, withdrawals and claims.

Oracle

Stale, future, incomplete, out-of-bound and excessive-deviation rounds.

Custody

Reentrancy, fee-on-transfer tokens, replay and duplicate claims.

Deployment Checklist

Bytecode

Deploy only byte-identical code bound to the printed scope root.

Configuration

Bind endpoints, production feeds, roles and exact sale window.

Funding

Fund AGNT inventory and refund reserves before purchase-capable modes.

Mainnet Acceptance

Flow

Execute controlled funded-wallet purchase, redemption and staking acceptance.

Evidence

Retain transaction, receipt, input, value, events, confirmations and canonical block hash.

Decision

Activation is a release gate, not an unresolved source finding.

Evidence Hashes

Scope root

5c51c8e5df9126e3b0a5ce15e98869f6130e43bb13e4caaebf30853286763f0c

Evidence run

20260730T020710Z

Package

Frozen manifest, coverage, Slither, triage and dependency evidence retain SHA-256 digests.

Toolchain Versions

Compiler

solc 0.8.24; optimizer 500; EVM paris.

Analysis

Hardhat 2.29.0; solidity-coverage 0.8.17; Slither 0.11.5.

Source File Register

Core

Five production core contracts.

Interfaces

Six production interfaces.

Package

Thirty-five frozen inputs including tests, scripts and configuration.

Oracle Risk Model

Inputs

BNB/USD, USDT/USD and USDC/USD feeds have independent age, bound and deviation settings.

Recovery

Risk parameters update only while purchases are paused.

Redemption Accounting

Basis

Wallet cost basis survives partial refunds and phase changes.

Replay

Each exact redemption consumes one wallet execution nonce.

Staking Custody

Entitlement

Position stores asset, principal, USD value, term, reward and maturity.

Release

Beneficiary proceeds release only after the matured AGNT claim.

Limitations

Point in time

The assessment applies only to the printed source identity.

External systems

Live RPC reliability, key custody, exchange operations and future upgrades are not guaranteed.

No warranty

Assessment cannot prove the absence of every defect.

Conclusion

ASSESSMENT COMPLETE

Code status

All nine tracked findings are resolved.

Verification

Compilation, tests, wrong-chain, coverage, static analysis and dependency checks completed.

Release

Mainnet activation remains conditional on operational acceptance gates.

Disclaimer

Technical scope

This document records a point-in-time smart-contract security assessment.

No investment advice

Nothing in this document is financial, legal, tax or investment advice.

No absolute assurance

Testing cannot guarantee that all vulnerabilities are absent or future changes remain secure.
